Discover: Enzyme Action: Preparing the pH Investigation

Today I can… ☐ Use these words in the right way. Use them to explain an idea: enzyme, substrate, active site and specificity. ☐ Say how an unsuitable pH can change the shape of an active site. Say how this could slow the enzyme's work. ☐ Name the things in the test: pH is what we change. It is the independent variable. Time until starch disappears is what we measure. It is the dependent variable. Choose suitable control variables. These are things to keep the same.

Feedback: ☐ VF ☐ WS ☐ I ☐ NS ☐ E ☐ R ☐ [//] ☐ [?] _____

Knowledge organiser

Learning goal

I can explain enzyme specificity and denaturation, then identify the variables and evidence needed to investigate the effect of pH on amylase safely.

A complementary active site

Specificity means an enzyme acts on particular substrates. Amylase breaks starch down into smaller sugars. It speeds this change up. The products leave. The enzyme is not used up. It can act again. This shape model is simple. It does not show real molecules as rigid shapes.

Connect pH to a measured endpoint

pH tells us how acid or alkaline a substance is. An unsuitable pH can change the active site. This can slow activity. Denaturation disrupts the enzyme's three-dimensional shape and how it works. Slow activity alone does not prove permanent denaturation.

Plan a valid comparison

Change pH. Time the first sampled drop that leaves iodine orange-brown. This means starch is not detected. Blue-black means starch is detected. Match temperature, concentrations, volumes and sampling. Check separate drops. Keep iodine out of the main mixture.

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Read the evidence honestly

A comparison with no active amylase should still test positive for starch. If no endpoint is seen by 300 s, write "no endpoint by 300 s". Do not make up a zero time or zero rate. The best pH is still to be found.

Diagram

Purple shows amylase. Yellow shows starch and smaller sugar products. The shape model shows two moments.

starch substrate

amylase enzyme

smaller sugar products

active site